



Occupational Exposure to Blood and Body Fluids Procedure

1. Applicability

Site	Operational Area	Applicable to
Armadale Kalamunda Group	All areas	All workforce groups

2. Purpose

This document describes the requirements for the management of healthcare workers (HCWs) who sustain an occupational exposure (OE) to blood or body fluids in a healthcare setting and are potentially exposed to blood-borne viruses.

3. Scope

An OE is defined as an incident that occurs during a HCWs work and involves direct contact with another person's blood or body fluids. Transmission of hepatitis B virus (HBV), hepatitis C virus (HCV) or human immunodeficiency virus (HIV) can occur via parental (skin penetration) or non-parental (mucosal or non-intact skin) exposure.

Generally, a HCW who sustains an OE has a low risk of contracting a blood-borne virus (BBV). The risk of transmission is dependent on the type of injury and extent of the exposure, and the current viral load of the source of the exposure. A thorough risk assessment of each incident is required to ensure appropriate management (refer to Appendix I).

Occupational exposure data is reported monthly to the Infection Prevention, Policy and Surveillance Unit (IPPSU), Department of Health.

4. Definitions

Term	Definition
BBVs	Blood borne viruses
EPP	Exposure Prone Procedure A subset of invasive procedures where there is potential for contact between skin of the healthcare worker and sharp surgical instruments, needles or sharp tissue in body cavities or in poorly visualised or confined areas of the body
HBV	Hepatitis B virus
HCV	Hepatitis C virus
HCW	Healthcare worker
HIV	Human immunodeficiency virus

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Term	Definition
Non-parental exposure	Contamination of mucous membranes e.g. eyes, mouth, non-intact skin with blood or body fluids
Non-responder HBV vaccine	A person without HBV infection who has a documented history of an age-appropriate primary course of hepatitis B vaccine, but with a HBsAb level <10 IU/mL
OE	Occupational exposure
Parental exposure	Piercing of skin or mucous membrane with a sharp that is contaminated with blood or body fluids
PEP	Post-exposure prophylaxis Administration of antiviral drugs, immunoglobulins or vaccines after exposure to a blood borne virus i.e. HIV or HBV in an attempt to prevent seroconversion
Recipient	The person who is exposed to another person's blood or body fluids
Seroconversion	A change in serological test results from negative to positive as antibodies develop in reaction to an infection or vaccine
Source	The person from whom blood or body fluids originated from
Serological window period	The time from exposure to seroconversion when the source may be asymptomatic, experiencing seroconversion illness, and when routine antibody testing may be negative

5. Exposure management

Any HCW who sustains an OE is to be managed in a prompt manner. In the event of an OE, the following procedure is required.

5.1 First aid and initial management

- Wash the wound or exposed skin thoroughly with soap and water or use an antiseptic wipe or skin cleanser. Apply a waterproof dressing as necessary and apply pressure if bleeding is still occurring. Do not squeeze or rub the injury site.
- Rinse eyes thoroughly (remove contact lenses), for at least 30 seconds, with water or normal saline. If blood or body fluids are sprayed into the mouth, spit out and then rinse the mouth with water several times.
- If any clothing is contaminated, remove and shower if necessary.
- Staff member should report incident to direct line manager/supervisor immediately.
- Complete the incident investigation report on [Combined Hazards or Incident Reporting \(CHoIR\)](#) and complete the forms in the [Occupational Exposure pack](#)

5.2 Incident Reporting and Management

- Obtain an "Occupational Exposure Pack" which are available in the following areas:

Armada Health Service (AHS)

- Infection Prevention and Management (IPM), Theatres, ED, Ward Nurses Station

Kalamunda Hospital (KH)

- Coordinators Office (Anderton), Nurses Station (DPU)

Report incident to:

- AHS and KH
 - In Hours (0700-1600) IPM Tel: 9391 2068
 - After Hours (1600-0700) After-Hours Nurse Manager (AHNM) via switchboard
- A risk assessment of the exposure will occur, if deemed **high risk** (see Appendix I) the RPH Immunology Registrar will be contacted via RPH Switchboard for advice on follow-up required
- Complete the incident investigation report on [Combined Hazards or Incident Reporting \(CHoIR\)](#)

5.3 Baseline blood-borne virus testing

- If serological testing is required, testing to be performed at a PathWest Collection Centre

NOTE: If after-hours, can be taken by Medical Officer otherwise the next day at PathWest.

- PathWest pathology request forms for both source and recipient are available in the Occupational Exposure Pack
- Where possible, the source must be identified. The medical team caring for the source patient should be notified prior to any baseline testing being performed.
- Medical staff must obtain informed consent for BBV testing from the source prior to performing any baseline testing (Table 1).
- Where the source is a neonate or an infant i.e., up to 6 months of age, it is preferable to collect the blood from the mother.

Table 1: Source and recipient serology baseline testing

Testing	Baseline tests required	Rationale
Source (all cases)	HBsAg, HIV Ag/Ab and HCV Ab	Evidence of disease
If source known positive HCV Ab	Add HCV RNA	Determine viral load/degree infectivity
If source known positive HBsAg	Add HBeAg and quantitative HBV DNA	
If indicated by risk assessment	Add syphilis serology ¹	
Recipient (all cases)	HBsAb, HIV Ag/Ab and HCV Ab ²	Evidence of HBV immunity and baseline record of results
If indicated by risk assessment	Add syphilis serology ¹	Baseline record of liver function
If source known HBV, HCV or HIV positive	Add liver function tests (LFTs)	
If recipient known non-responder	Add HBsAg, HBcAb	Evidence of pre-existing HBV infection

1. Given increasing rates of syphilis in WA, consider testing the source and recipient as indicated by history and risk assessment (e.g within obstetric units).
2. A positive HCV Ab should be followed up with a HCV RNA PCR test.

6. Management of source following baseline blood-borne virus testing

Table 2 outlines the management of OEs dependant on the source status.

Table 2: Management of source based on baseline testing results

Source results	Management
Negative for BBV	• If negative for HBV, HCV and HIV, further testing not required unless there is a reason to suspect the source was involved in recent high-risk behaviours for BBV infection. • Follow-up can be undertaken through source's general practitioner (GP) if required
Positive for BBV	• Pre-test counselling should include the need for further testing should a source return a positive result. • If not already in the care of an appropriate medical specialist, referral by the treating team is required.
Likely to be positive for BBV	• If suspected that the source is in the "serological window period" for a BBV, provide appropriate counselling. • Seek consent to follow up source at appropriate intervals i.e., 6 weeks and 12 weeks, to ascertain evidence of disease

7. Management of recipient following baseline blood-borne virus testing

- Upon receipt of baseline testing results, post-test counselling and test results will be discussed with the recipient.
- If the recipient is found to be infected with a BBV and not already in the care of an appropriate medical specialist, they will be referred to the RPH Immunology service as soon as possible.
- Recipients may opt to attend their own GP for follow-up.
- If the source is negative on baseline testing for all BBVs, the recipient should be offered follow-up testing at 3 months for reassurance. No further follow-up of source is required. No behavioural or work practice modifications are required by the HCW.
- If there is a high risk of the source being infected with a BBV, then the HCW is to be managed in accordance with a source positive approach.

8. BBV-specific management of recipient for a positive or likely positive source

8.1 Source positive for HBV or likely to be positive.

- If the recipient has a documented protective response i.e., anti-HBs level $\geq 10\text{IU/L}$, at any time following completion of a vaccination course, then no further action is required regardless of exposure.
- If recipient considered non-immune and the source is positive or potentially positive for HBV, manage as per Appendix II.
- The recipient should receive follow-up testing as per Table 3.

8.2 Source positive for HCV or likely to be positive.

- Currently there is no prophylaxis proven to be effective in altering the likelihood of HCV transmission.
- Any recipient exposed to an HCV-positive source will be reviewed and counselled by RPH Immunology or AKG Infectious Diseases Consultant via switchboard
- The recipient is to have follow up testing as per Table 3.
- The recipient should be advised that during the follow-up period they should refrain from donating plasma, blood, organs, body tissue, breast milk or sperm. The recipient is not required to modify sexual practices or refrain from becoming pregnant or breastfeeding.
- No modifications to a recipient's patient care responsibilities are required based solely on exposure to HCV positive blood, however those recipients who perform EPPs may require more frequent testing.
- The recipient is to be advised to seek medical attention if they become unwell with symptoms consistent with acute hepatitis such as nausea, vomiting, abdominal discomfort or jaundice.
- Ongoing support must be provided for the duration of post-exposure follow-up and extended to the recipient's significant others as required.

8.3 Source positive for HIV or likely to be positive.

- Any recipient exposed to a known HIV positive source is to be referred immediately to the RPH Immunologist on 9224 2244 for consideration of initiation of HIV PEP.
- The decision to commence HIV PEP is based on the type of exposure and the risk associated with that exposure, source characteristics such as stage of HIV infection, viral load and antiretroviral treatment history.
- If HIV PEP is indicated, it should be commenced as soon as possible following exposure, preferably within 1-2 hours and no longer than 72 hours.
- Irrespective of decision to take HIV PEP, or the type of exposure, the recipient is to have follow-up testing as per Table 3.
- During the follow-up period the recipient should be advised:
 - Refrain from donating plasma or blood for a period of 12 months
 - Refrain from donating body tissue, breast milk or semen
 - Exercise sexual abstinence or use condoms to protect sexual partners and avoid pregnancy.
 - Not share razors, toothbrushes or other possible sources of BBV transmission
 - Cover open cuts and wounds with a waterproof dressing.

- No modifications to a recipient's patient care responsibilities are required based solely on exposure to HCV positive blood, however those recipients who perform EPPs may require more frequent testing.
- Ongoing support must be provided for the duration of post-exposure follow-up and extended to the recipient's significant others as required.

Table 3: Recipient follow-up testing recommendations

Source	Follow-up testing recommendations
HBV positive and non-immune recipient	• LFTs at 6 weeks and 12 weeks • HBsAg at 12 weeks and 24 weeks (may give a false positive if tested within 2 weeks of giving hepatitis B vaccine) • HBsAb at 4-6 weeks post vaccination or delay for 4-5 months if HBIG administered
HCV positive	• HCV RNA PCR at 4, 8 and 12 weeks post exposure • HCV antibody at 12 weeks and 24 weeks
HIV positive	• HIV antibodies at 4-6 weeks and 12 weeks post exposure

9. Compliance monitoring

Compliance against this procedure will be evaluated against monthly surveillance data, information provided by the Department of Health and CHOIR reports. Occupational exposure data is reported monthly to the Infection Prevention, Policy and Surveillance Unit (IPPSU), Department of Health.

10. Legislative context

The following documents are required to give effect to this policy:

- *Work Health and Safety Act 2020*
- WA Department of Health [Health Care Worker Immunisation Policy](#)
- EMHS [Infection Prevention and Management Policy](#)

11. Supporting information

The following documents support the implementation of this policy:

- AKG [Healthcare Worker Immunisation and Health Policy](#)
- WA Department of Health [Management of Occupational Exposure to Blood or Body Fluids in Healthcare Settings](#)

12. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 3 – Action 3.15 Preventing and Controlling Infections

13. References

1. Department of Health, Western Australia. (2024). *Management of Occupational Exposure to Blood or Body Fluids in Healthcare Settings*.
2. Communicable Diseases Network Australia (CDNA). (2019). *Australian national guidelines for the management of healthcare workers living with blood borne viruses and healthcare workers who perform exposure prone procedures at risk of exposure to blood borne viruses*.
3. Department of Health, Western Australia. (2022). *Healthcare infection surveillance of Western Australia (HISWA) Surveillance manual*.
4. Australian Technical Advisory Group on Immunisation (ATAGI). (2023). Post-exposure prophylaxis for non-immune people exposed to a source that is positive for hepatitis B surface antigen or has an unknown status. *The Australian Immunisation Handbook*. Commonwealth of Australia: Department of Health and Aged Care. [Table. Post-exposure prophylaxis for non-immune people exposed to a source that is positive for hepatitis B surface antigen or has an unknown status | The Australian Immunisation Handbook \(health.gov.au\)](#)
5. National Health and Medical Research Council (NHMRC). (2019). *Australian Guidelines for the Prevention and Control of Infection in Healthcare*

14. Policy contact

Enquiries relating to this policy may be directed to:

Title: Clinical Nurse Specialist
 Department: Infection Prevention & Management
 Email: ahsinfection.controlnurses@health.wa.gov.au

15. Review

This policy will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V5.01	18/02/2025	18/02/2028	Scheduled review

16. Authorisation

This policy has been authorised and issued by Director Nursing and Midwifery:

Authorised by	Denver Prince – A/Director Nursing and Midwifery
Authorisation date	14/02/2025
Published date	18/02/2025

Appendix I: Risk assessment and classification of occupational exposures

Risk	Type of injury
High Risk	• Transfusion of blood. • Injection of large volume of blood or body fluids (>1ml). • Parenteral exposure to laboratory specimens containing high titre of virus.
Moderate Risk	• Skin penetrating injury with a needle contaminated with blood or body fluid. • Injection of blood or body fluid < 1ml. • Laceration or similar wound which causes bleeding and is produced by an instrument that is visibly contaminated with blood or body fluid. In laboratory settings, any direct inoculation with material likely to contain HIV, HBV or HCV.
Low Risk	• Intradermal (superficial) injury with a needle contaminated with blood or body fluid. • A wound not associated with visible bleeding, caused by an instrument contaminated with blood or body fluid. • Prior wound or skin lesion contaminated with blood or body fluid. • Mucous membrane or conjunctival contact with blood or body fluid. • Scratched / broken skin caused by a fingernail injury when there is blood evident on the source hands. • Human bites that break the skin - the clinical evaluation must include the possibility that both the person bitten and the person who inflicted the bite were exposed to BBVs.
Very Low Risk	• Intradermal (superficial) injury with a needle considered not to be contaminated with blood or body fluid. • Superficial wound not associated with visible bleeding, caused by an instrument considered not to be contaminated with blood or body fluid. • Prior wound or skin lesion contaminated with blood or body fluid. • Mucous membrane or conjunctival contact with a body fluid other than blood.
No Risk	• Intact skin visibly contaminated with blood or body fluid. • Needlestick with non-contaminated i.e clean needle or sharp

Appendix II: Recommended HBV post-exposure prophylaxis (PEP)

Recipient Status	Source HBsAg Positive or Unknown or Unable to be Tested
Unvaccinated	Administer HBIG* as a single dose within 72 hours of exposure and initiate hepatitis B vaccination within seven days and at one and six months after 1 st dose.
Previously vaccinated but known non-responder**	Administer HBIG* as a single dose within 72 hours of exposure. Offer further HBV vaccination doses and follow-up with HBsAg as per the Australian Immunisation Handbook
Previously vaccinated but response unknown / vaccination incomplete.	If HBsAb < 10 IU / L administer HBIG* as a single dose within 72 hours of exposure and initiate hepatitis B vaccination within seven days. If HBsAb ≥ 10 IU / L no treatment is required.
Previously vaccinated Known responder with documented HBsAb level ≥ 10 IU / L at any time post vaccinations.	No Treatment required.
Known HBV positive	Persons previously infected with HBV do not require PEP.
*Dose of HBIG - 400IU by IM (100IU if body weight < 30kg). HBsAb response must be done when passively acquired antibody from HBIG is no longer detectable i.e. four-five months. **Non-responder - is a person without HBV infection who has a documented history of an age-appropriate primary course of hepatitis B vaccine, but with a current HBsAb level < 10 IU / ml. There are a number of potential options for non-responders. Persons who do not respond to the primary vaccination course, and in whom chronic HBV infection has been excluded, must be offered further HBV vaccination doses. (Refer to The Australian Immunisation Handbook). ***Review vaccination history and administer additional doses of HBV vaccine at one month and months after 1st dose if required. Re-test for HBsAb 4-6 weeks post course completion.	